Introduction to Artificial intelligence

Tutorial Sheet 7 (Logic and Inference - Solution)

Exercise 1

A. Rewrite the following statements in First-Order Logic. You may use appropriate predicates. The used predicates should reasonably and logically make sense.

i. Every coyote chases some roadrunner.
ii. Every roadrunner who says ``beep-beep'' is smart.
iii. No coyote catches any smart roadrunner.
iv. Any coyote who chases some roadrunner but does not catch it is frustrated.

B. Convert the statements in CNF showing all the steps.

i. $\forall x\ (COYOTE(x) \Rightarrow \exists y\ (RR(y) \wedge CHASE(x,y)))$
ii. $\forall x\ (RR(x) \wedge BEEP(x) \Rightarrow SMART(x))$
iii. $\neg \exists x \exists y\ (COYOTE(x) \wedge RR(y) \wedge SMART(y) \wedge CATCH(x,y))$
iv. $\forall x\ (COYOTE(x) \wedge \exists y\ (RR(y) \wedge CHASE(x,y) \wedge \neg CATCH(x,y)) \Rightarrow FRUSTRATED(x))$

Exercise 2 (Taken from Tutorial 6):

A. Rewrite the following statements in First-Order Logic. Choose appropriate predicates and constants.

1. All dogs howl at night.
2. Anyone who has any cats will not have any mice.
3. Light sleepers do not have anything which howls at night.
4. Tariq has either a cat or a dog.
5. (Conclusion) If Tariq is a light sleeper, then Tariq does not have any mice.

B. Convert the statements in CNF showing all the steps.

C. Apply the Resolution inference rule to prove the conclusion (statement 5) from the first four statements (statements 1 to 4). Make sure that each step all the substitutions are shown as well as the parent clauses and the resolvent.

Exercise 3:

1. Translate the following sentences into for the First-Order Logic sentences with an appropriate encoding:

a) There are three children Ali, Boudour and Laith.
(Hint: you must explicitly ensure that different names refer to distinct individuals).

b) Every child is wearing either a hat, coat or a scarf.
c) Ali is wearing a red hat.
d) Boudour is wearing a green coat.
e) Laith is wearing a blue scarf.
f) Ali has a favourite colour, which is either red, green, or blue.
g) Laith's favourite colour is green.
h) Boudour's favourite colour is red.